LAB REPORT

Data Exfiltration Detection

Detecting Unauthorized Data Transfers Across Network Channels

Blue Team, SOC Level 1, Zen Mier

1. Objectives

Data exfiltration is when an attacker who has already broken into a network secretly moves sensitive files or information out of the organization. This is often the final and most damaging step of an attack. This lab covers how attackers use common, normally allowed network protocols to smuggle data out, and how to detect it.

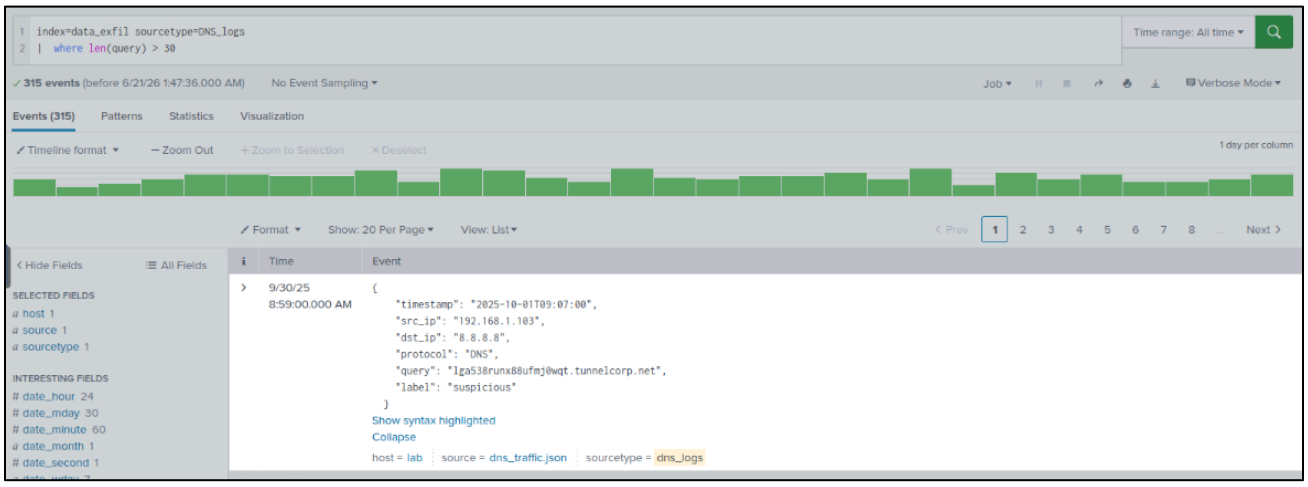
- Understand how data exfiltration works and why it is difficult to detect when attackers use legitimate-looking protocols.
- Detect data being hidden inside DNS traffic, a technique called DNS tunneling.
- Investigate FTP transfers to identify files being sent to suspicious external destinations.
- Analyze HTTP traffic for data being uploaded out of the network through web requests.
- Examine ICMP (ping) packets for abnormally large payloads carrying hidden data.

2. What Was Done

DNS Tunneling Detection

DNS is the protocol that translates website names into IP addresses. It is almost always allowed through firewalls because so many legitimate applications depend on it. Attackers exploit this by encoding stolen data inside DNS query strings, making it look like normal name lookup traffic.

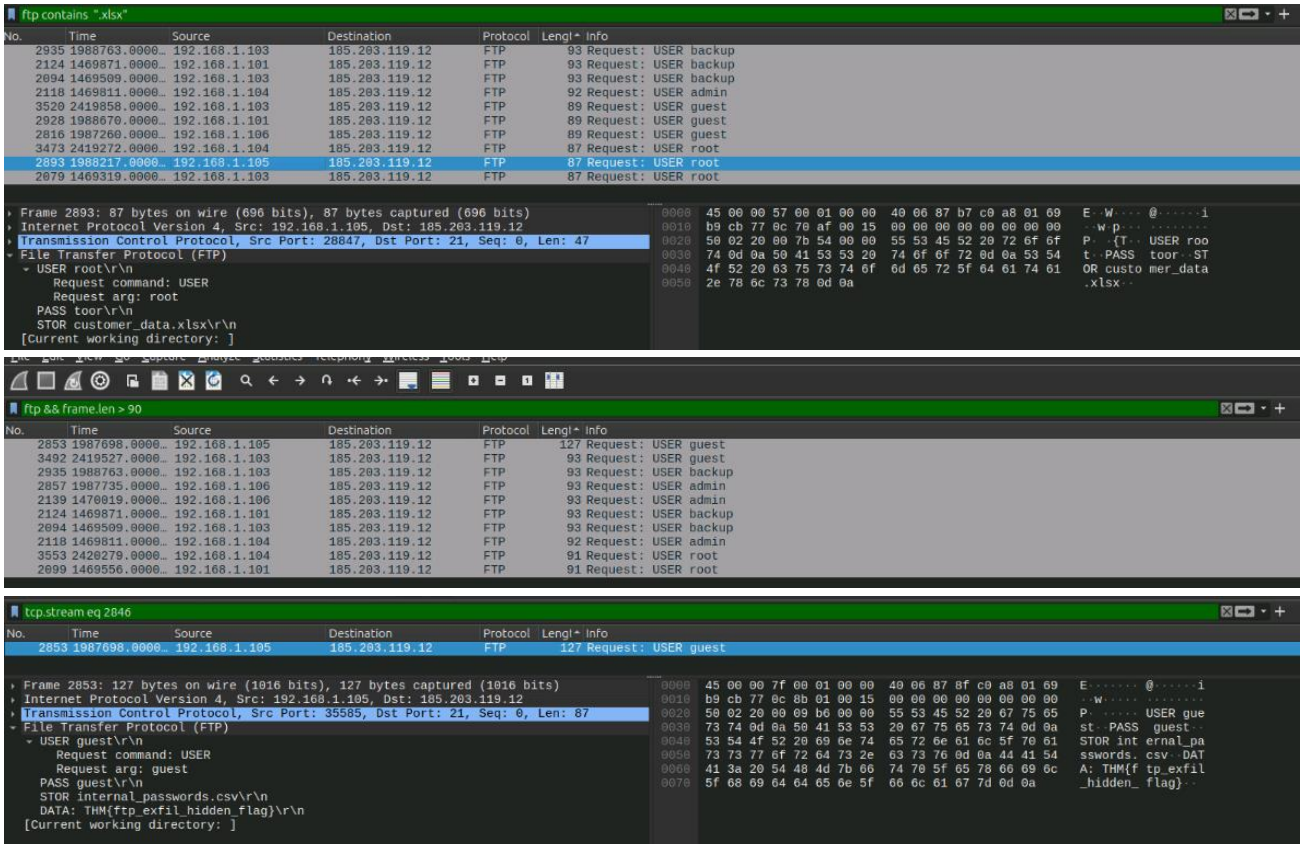
Using Splunk, DNS logs were filtered for abnormally long query strings and unusually high volumes of queries to the same external domain. A total of 315 suspicious DNS entries were found, all pointing to the domain tunnelcorp.net. The internal machine responsible for sending the most queries was identified as 192.168.1.103.



FTP Exfiltration Detection

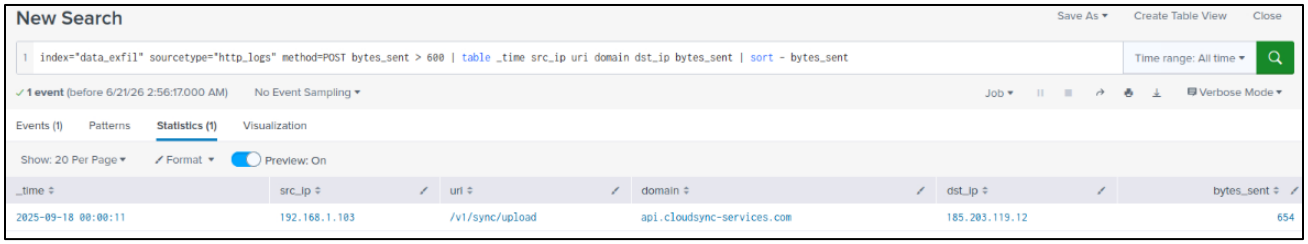
FTP (File Transfer Protocol) is used to move files between machines. Because it transmits data in plain text, everything including login credentials and file names, can be read directly from the traffic capture.

Wireshark filters were applied to isolate FTP sessions, revealing a suspicious transfer of a file named customer_data.xlsx being sent from an internal machine to an external IP. 5 connections were observed from guest account. The internal IP sending the largest payload was 192.168.1.105.



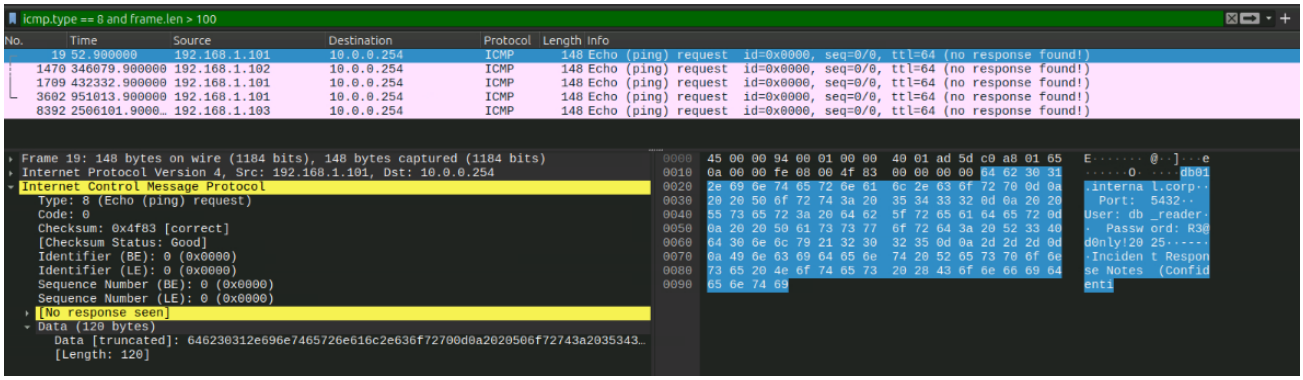
HTTP Exfiltration Detection

HTTP POST requests are normally used to submit forms on websites, but attackers can use the same method to upload files to a server they control. Splunk was used to filter for outbound POST requests with unusually large payloads, indicating data being pushed out rather than typical browsing behavior. This falls under the network-based exfiltration technique category.



ICMP Exfiltration Detection

ICMP is the protocol behind the simple ping command, used to check if a machine is reachable. A standard ping packet is very small (around 74 bytes). By filtering for ICMP packets with a frame length over 100 bytes in Wireshark, packets were identified that were carrying hidden data in the payload field, far more than any legitimate ping would contain.



3. Summary

This lab demonstrated how attackers abuse trusted, everyday protocols to move stolen data out of a network without triggering obvious alarms. The key lesson is that no single alert tells the whole story — effective detection requires correlating host logs, network traffic, and application logs together to spot the pattern of exfiltration across different channels.

Area Covered	Key Takeaway
DNS Tunneling	315 suspicious DNS queries to tunnelcorp.net detected, data encoded in query strings.
FTP Exfiltration	customer_data.xlsx transferred from 192.168.1.105 to an external destination.
HTTP Exfiltration	Outbound POST requests with large payloads detected network-based exfiltration technique.
ICMP Exfiltration	ICMP packets over 100 bytes identified carrying hidden data in echo request payloads.